STUDENT TASK & STUDY PLANNER

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Software Designing & Algorithms — Project Documentation

Language	GUI	Storage	Architecture
C#	Windows Forms	JSON	3-Layer

1. Introduction

Student Task & Study Planner is a basic desktop application designed to help students organize academic tasks and study activities. The application provides a simple dashboard, task management, study scheduling, searching, sorting, completion tracking, and JSON-based data storage. The project was intentionally kept at a manageable level so that its architecture, algorithms, and implementation can be explained clearly during demonstration and viva.

2. Problem Statement

Students often manage assignments, quizzes, projects, and study activities using scattered notes or separate applications. This project provides one small application where academic tasks can be recorded, viewed, searched, sorted, completed, and deleted. A dashboard gives a quick view of total, pending, and completed tasks.

3. Objectives

- Create a simple and user-friendly desktop task planner.
- Apply a 3-layer architecture to separate presentation, business logic, and data handling.
- Store task information permanently using a JSON file.
- Demonstrate Linear Search and Bubble Sort as project algorithms.
- Provide a presentable GUI suitable for a student-level software project.

4. Main Features

Feature	Purpose
Dashboard	Displays task statistics, completion progress, and recent tasks.
Add Task	Creates a task with title, description, due date, priority, and status.
View Tasks	Displays stored tasks and provides task-management actions.
Study Schedule	Provides a simple place to manage study-related scheduling.
Complete / Delete	Updates task status or removes a selected task.
Search / Sort	Searches task titles and sorts tasks by due date.

5. 3-Layer Architecture

The application is divided into clear layers. The Presentation layer handles the Windows Forms interface. The Business Logic layer contains task operations and algorithms. The Data layer handles reading and writing task data to the JSON file. The Models folder contains the Task object shared between these layers.

Presentation Layer	MainForm and other forms — buttons, tables, input, and user interaction.
Business Logic Layer	TaskManager — add, complete, delete, search, and sort operations.
Data Layer	TaskRepository — loads and saves tasks in DataFiles/tasks.json.
Models	Task class — stores title, description, due date, priority, and status.

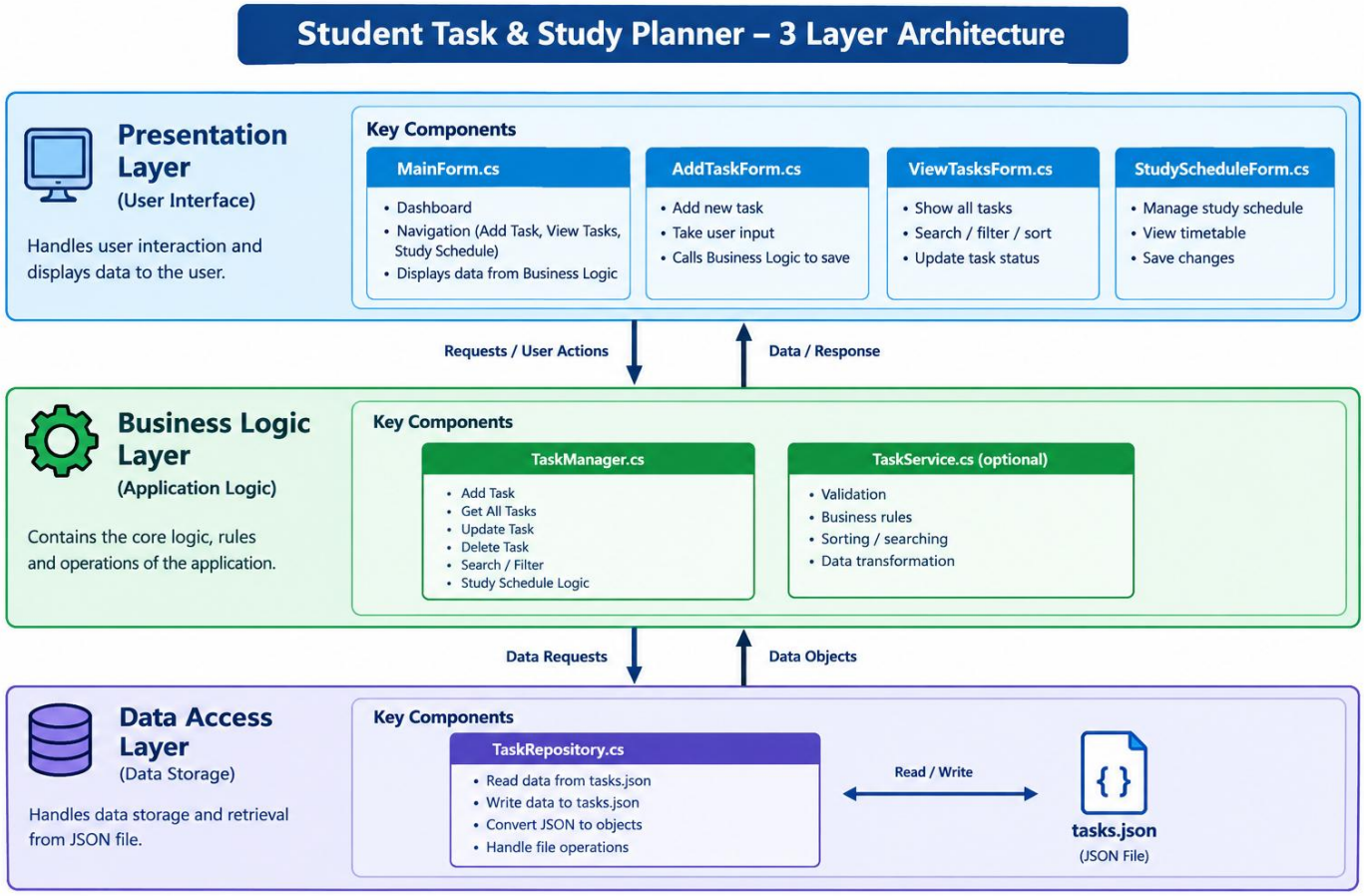


Figure 1: Detailed 3-Layer Architecture of the Student Task & Study Planner

IMPLEMENTATION, ALGORITHMS & TESTING

6. Project Structure

The project is organized into folders according to responsibility. This keeps the application easier to understand, test, and maintain.

```
StudentTaskPlanner
├── BusinessLogic
│   └── TaskManager.cs
├── Data
│   └── TaskRepository.cs
├── DataFiles
│   └── tasks.json
├── Models
│   └── Task.cs
├── Presentation
│   └── MainForm.cs
├── Program.cs
└── StudentTaskPlanner.csproj
```

7. Algorithms Used

Linear Search

Linear Search is used when searching for a task by title. The algorithm checks tasks one by one from the beginning until a matching title is found or the list ends. For n tasks, the worst-case time complexity is O(n).

Bubble Sort

Bubble Sort is used to arrange tasks according to their due dates. Adjacent tasks are compared and swapped when they are in the wrong order. For n tasks, its worst-case time complexity is $O(n^2)$.

8. Data Storage

Task information is stored in DataFiles/tasks.json. JSON was selected because it is lightweight, easy to read, and sufficient for a basic desktop project without introducing database-server complexity. The repository class handles loading and saving the task list, keeping file operations separate from the user interface.

9. Workflow

- User opens the application and views the dashboard.
- User selects Add Task and enters task information.
- The Presentation layer sends the task to TaskManager.
- TaskManager applies the required operation and calls TaskRepository.
- TaskRepository saves the updated list to tasks.json.
- The dashboard and task views refresh and display the updated information.

10. Testing Performed

Test	Result
Application startup	Application opens successfully and dashboard loads.
Add Task	Task information is accepted and stored.
View Tasks	Stored tasks are displayed correctly.
Complete Task	Selected task changes to Completed.
Delete Task	Selected task is removed from the list.
Search	Matching task titles are returned.
Sort	Tasks are arranged by due date.
Study Schedule	Schedule interface opens and functions.
Dashboard	Statistics and recent tasks update correctly.
Data persistence	Tasks remain available after restarting the application.

11. Conclusion

The Student Task & Study Planner demonstrates how a small C# desktop application can combine GUI design, layered architecture, file-based data storage, and basic algorithms. The final application was tested through its main buttons and workflows, and the interface was refined to provide a cleaner and more presentable student-level project.

Note: Project screenshots can be inserted in the final submission copy if the instructor requires visual evidence of the working application.